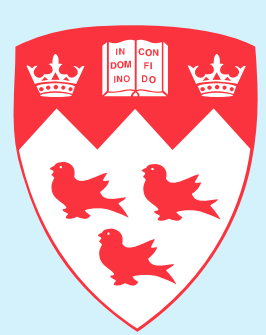


DCX USES DISTINCT PATHWAYS TO MODULATE MICROTUBULE DYNAMICS

Sofia Cruz Tetlalmatzl, Adam G. Hendricks, Gary J. Brouhard

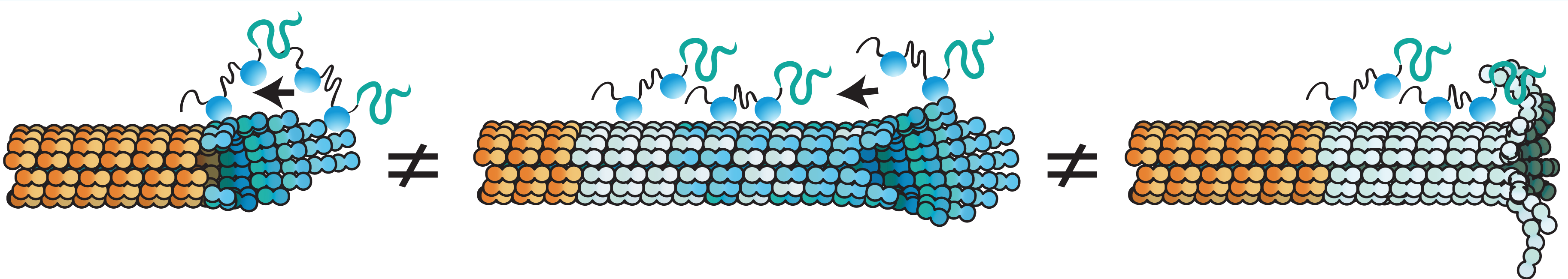
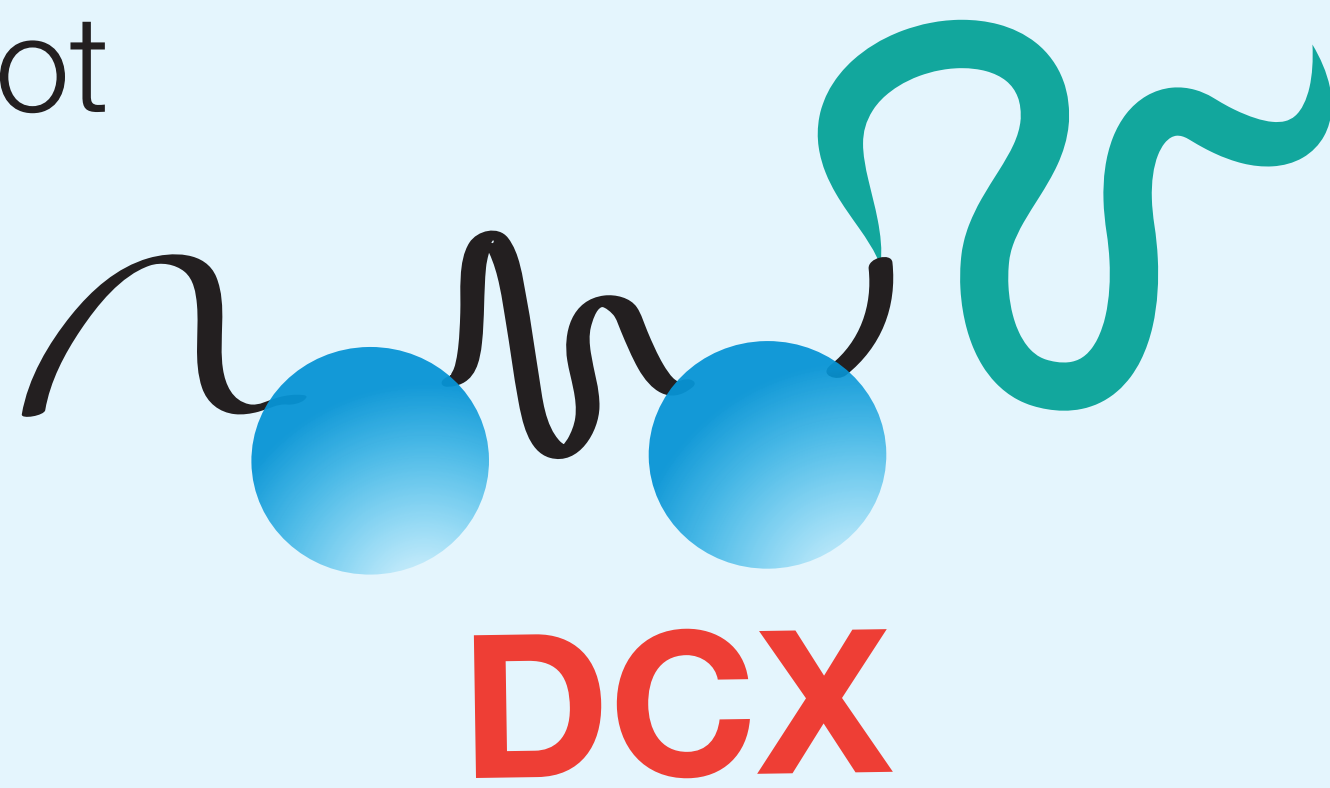


McGill

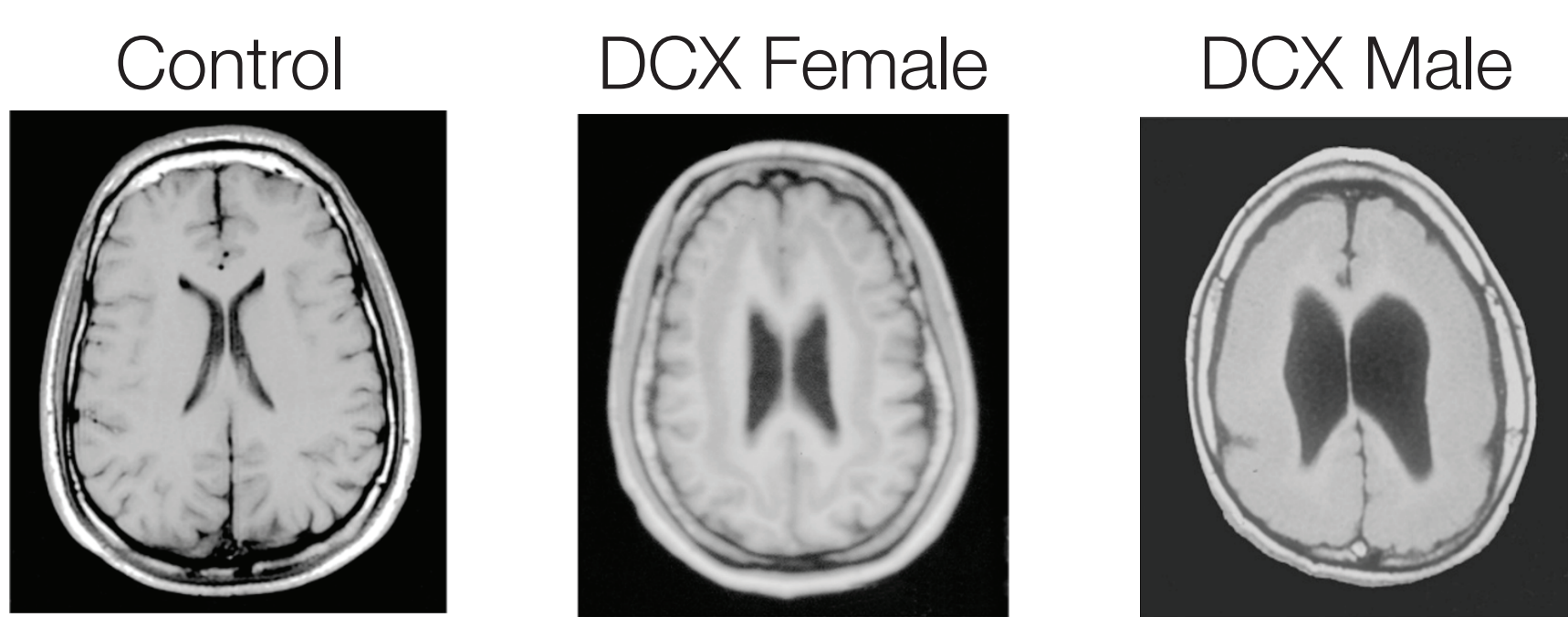


Take-home message

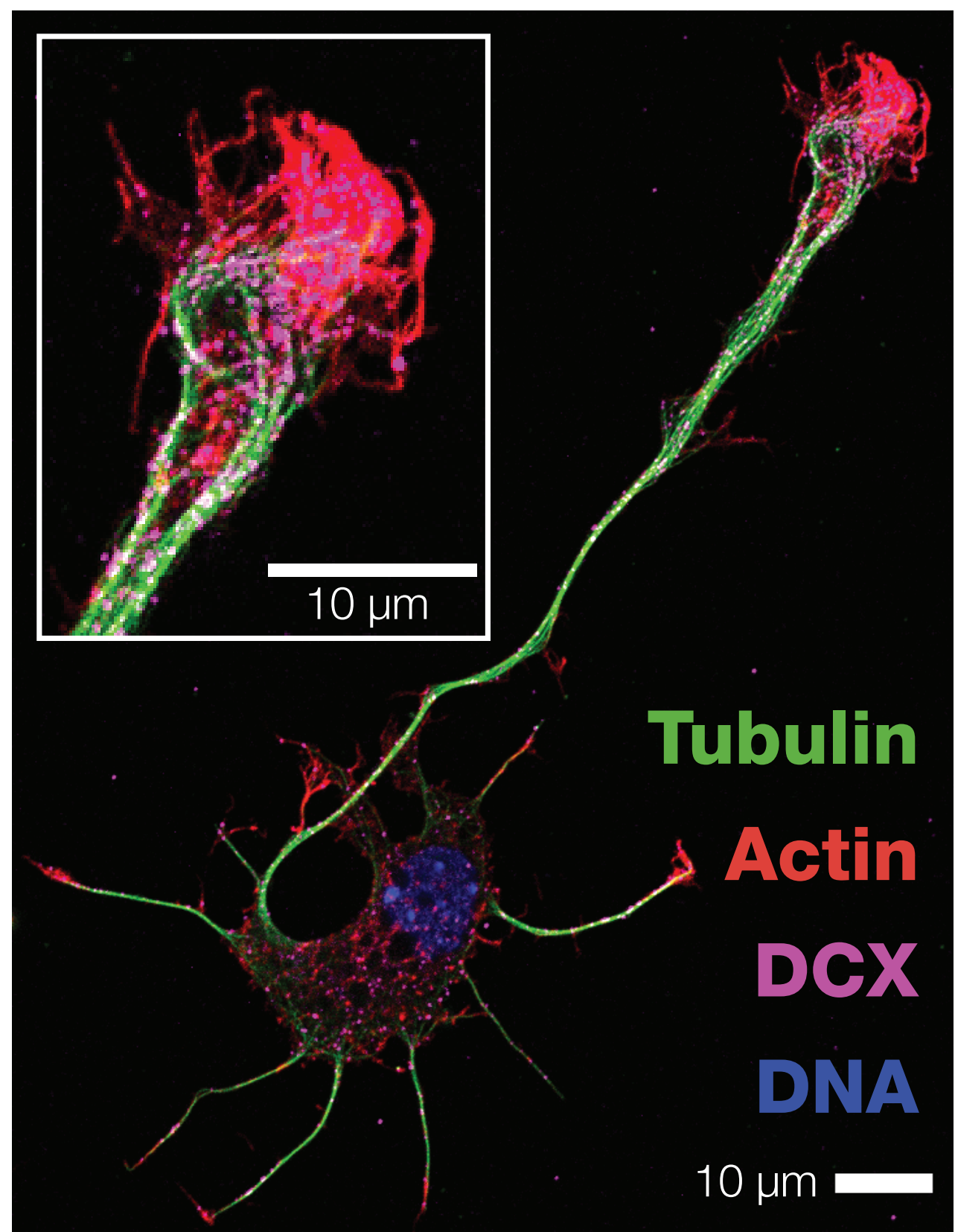
- Enhanced microtubule elongation is not necessary for accelerated nucleation
- Accelerated nucleation and slow shrinkage are not sufficient to induce rescues



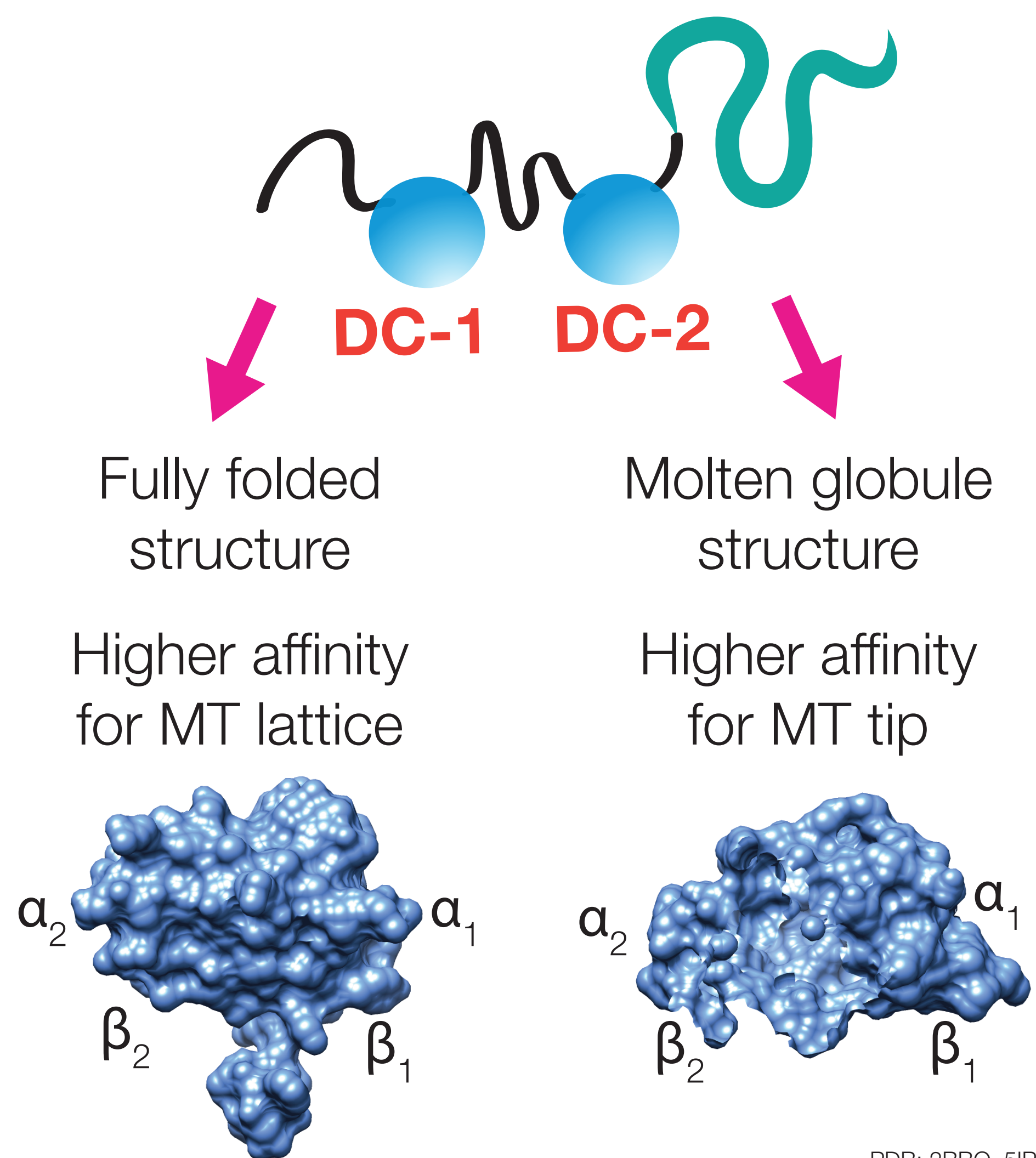
DCX Modulates Growth Cone Microtubule Dynamics



Feng 2001



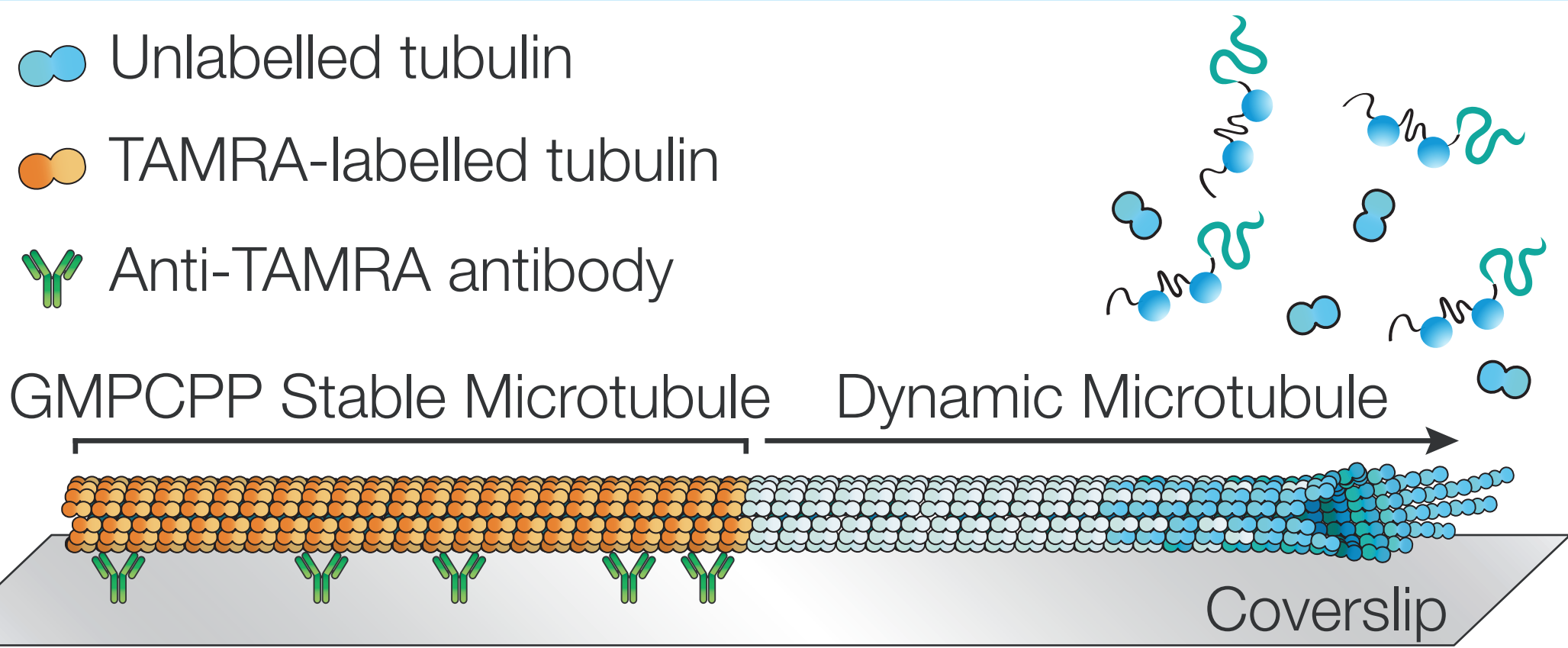
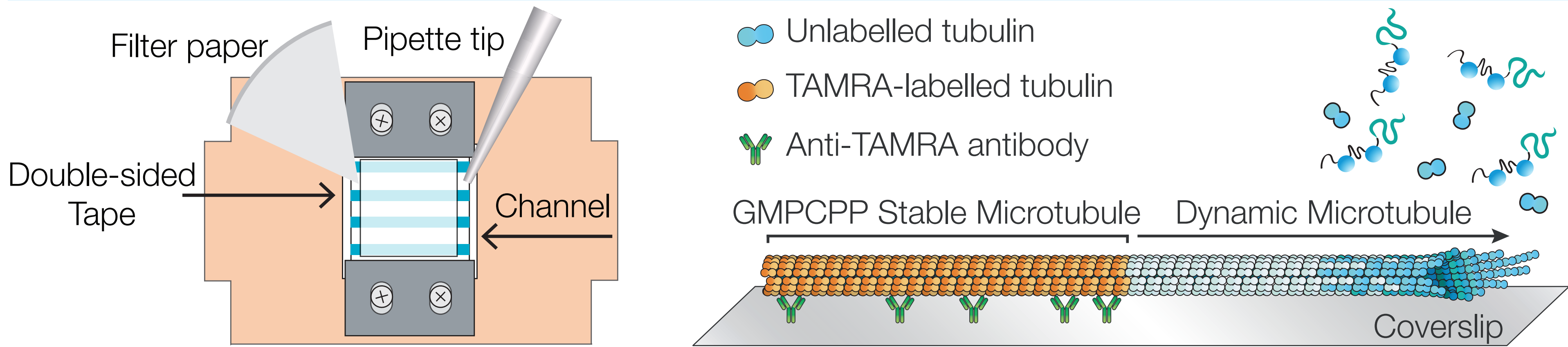
Muriel Sebastian



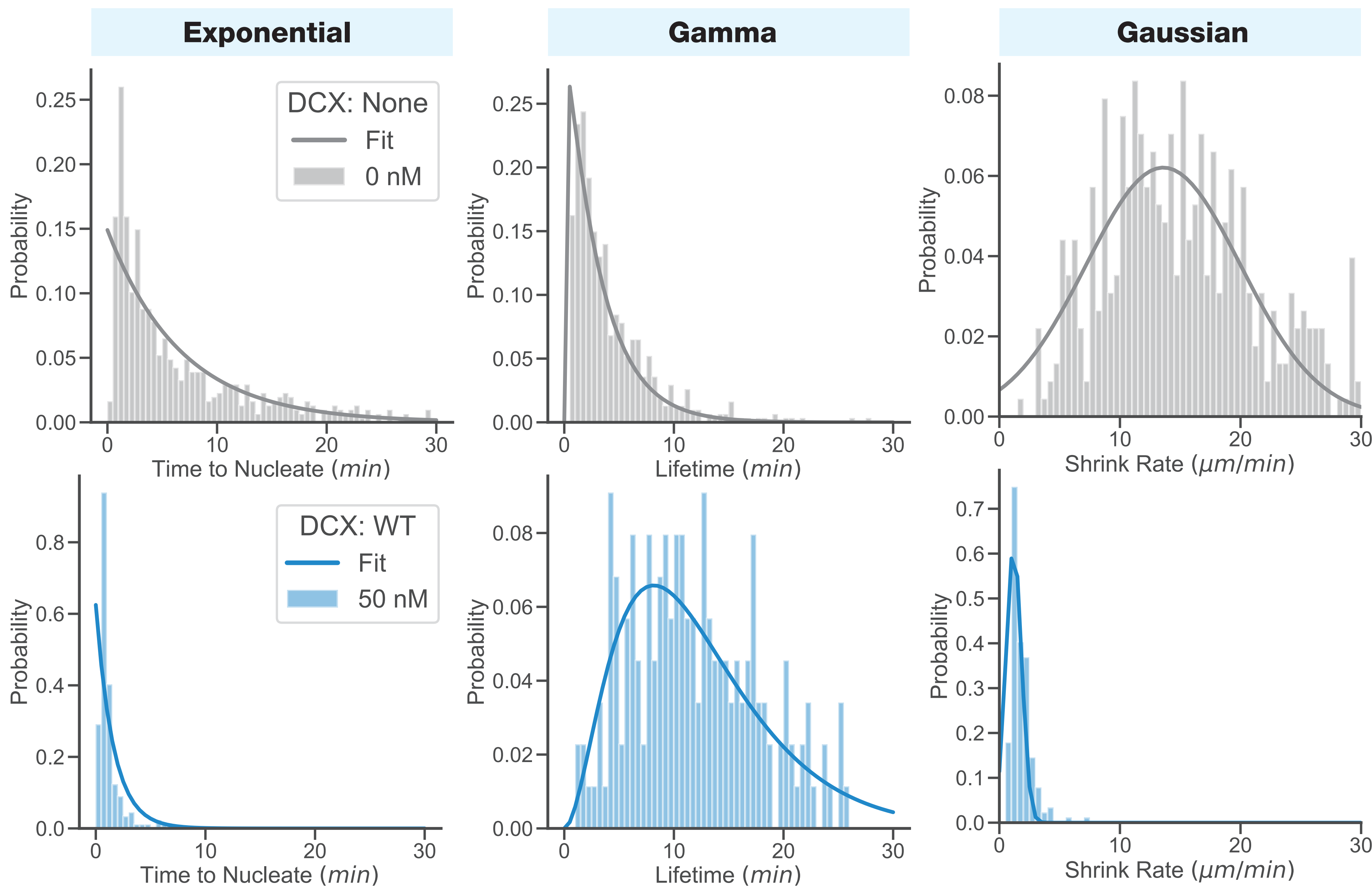
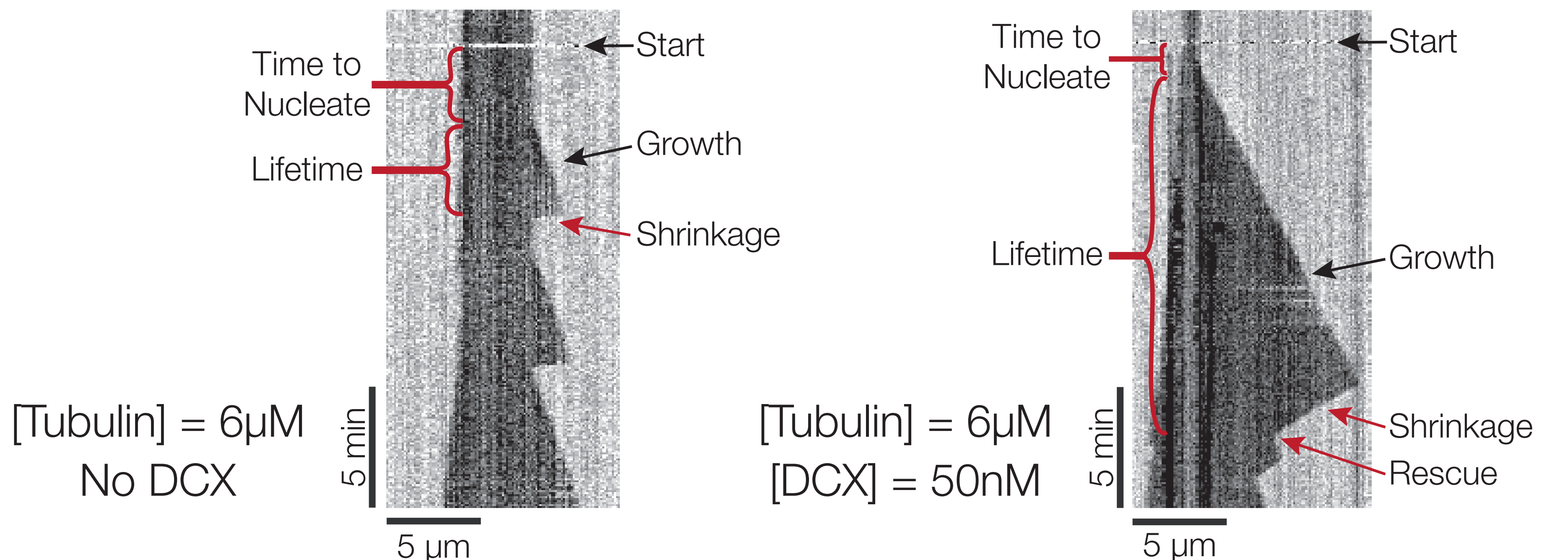
Difference in Function?

PDB: 2BBQ, 5IP4

in vitro Reconstitution Assay



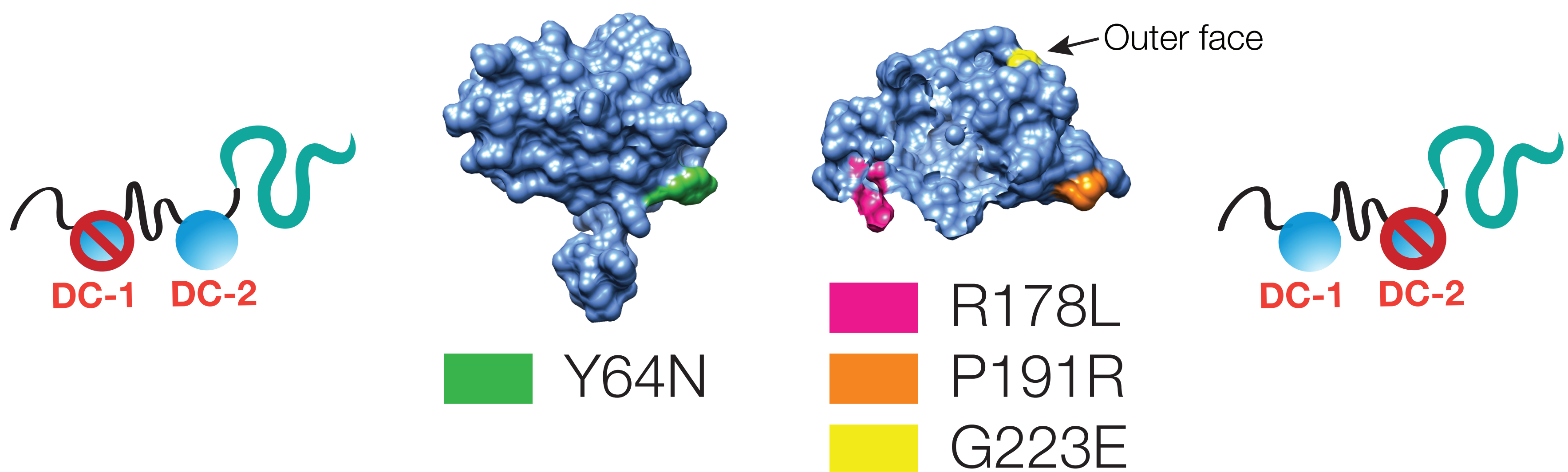
DCX Modulates *in vitro* Microtubule Dynamics



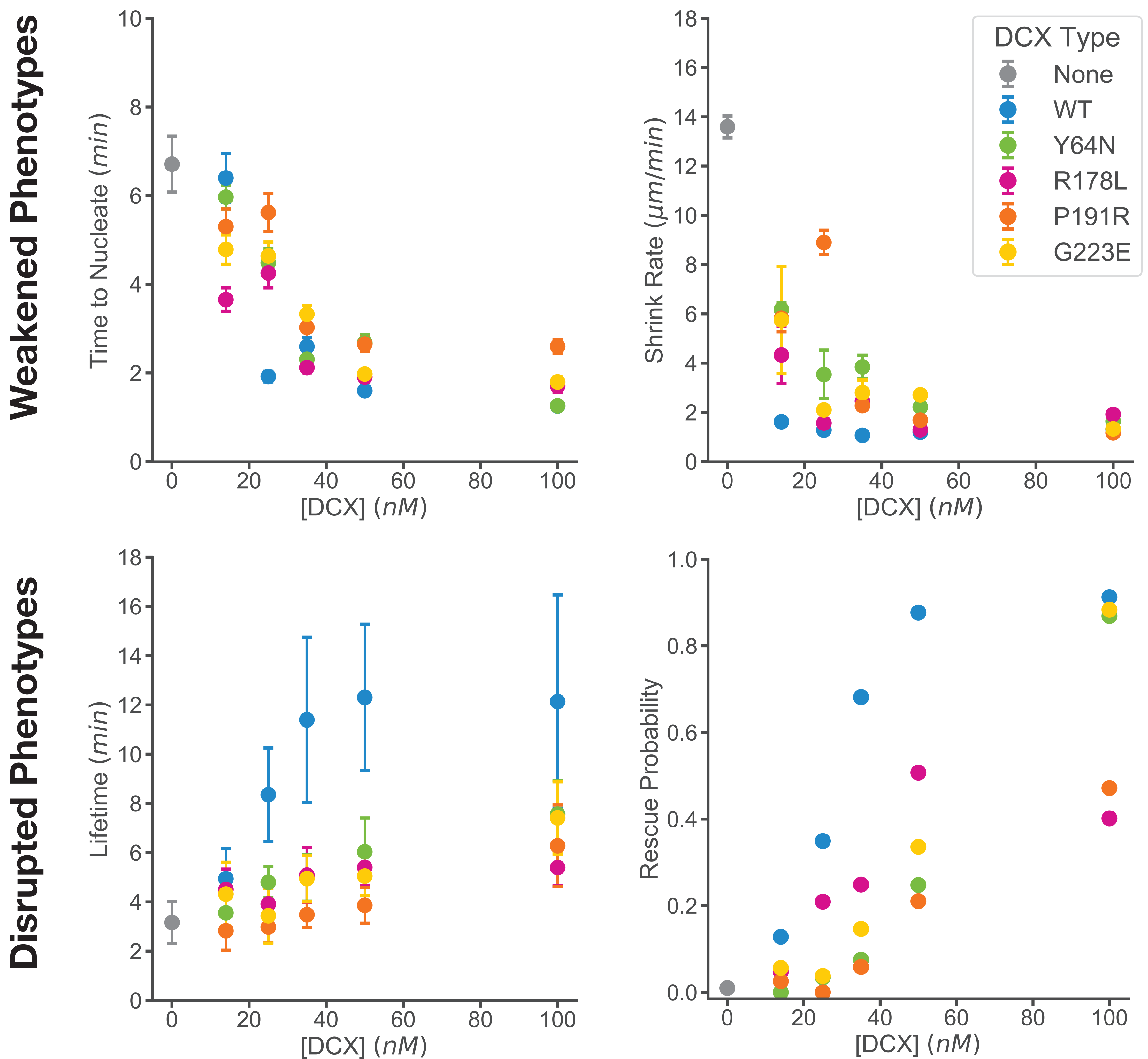
None: n = 615, WT: n = 179

Hypothesis

Patient mutations inhibit either domain



Patient Mutations Compromise Modulation



None: n = 615, WT: n = 281,229, 179, 179,160, Y64N: n = 209, 260, 306, 230, 206, R178L: n = 445,258,209,270,234, P191R: n = 429, 209, 222, 209, 142, G223E: n = 266,212,246,247,215, Error Bars: Fit SE, All data points have at least three repeats

Future Prospects

- Is enhanced elongation given by sequential or simultaneous domain binding?
- Are rescues driven by DCX-2's higher affinity for the MT tip?

References

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Thursday, February 25, 2021

12:00 PM – 1:30 PM

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